

Are there sex differences in placebo analgesia during visceral pain processing? A fMRI study in healthy subjects

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Key Messages

- We explored sex differences in the neural mechanisms underlying placebo analgesia for visceral pain.
- In an established deceptive placebo analgesia paradigm designed to induce positive (placebo) vs neutral (control) expectations, $N = 15$ healthy men and $N = 15$ healthy women were compared with respect to pain ratings and neural activation during expectation and delivery of painful rectal distensions.
- During visceral placebo analgesia, men and women differed in neural modulation during pain anticipation and painful rectal stimulation despite similar placebo-induced reductions in subjective pain evaluation.
- These findings in a small sample suggest a role of sex in the central mechanisms mediating cognitive pain modulation. More research in larger samples is needed to clarify the role of sex and/or gender in the central mechanisms contributing to chronic abdominal pain and pain-related treatment responses conditions such as the irritable bowel syndrome.

Abstract

Background We explored sex differences in the neural mechanisms mediating placebo analgesia in an established visceral pain model involving painful rectal distensions in healthy volunteers. **Methods** $N = 15$ men and $N = 15$ women underwent three consecutive functional magnetic resonance imaging sessions during which cued painful rectal distensions were delivered. After an adaptation session, positive expectations were induced with deceptive instructions regarding administration of an analgesic drug (placebo session). In the other session (control), truthful information about an inert substance was given. Sex differences in placebo-induced modulation

of neural activation during anticipation and pain were analyzed along with ratings of expected and perceived pain intensity. **Key Results** Placebo-induced reductions in pain ratings were comparable between men and women. At the level of the brain, group comparisons with respect to differences between the placebo and control conditions revealed greater modulation of the posterior insula (regions-of-interest analysis: $p_{FWE} < 0.05$) and dorsolateral prefrontal cortex (whole-brain analysis: $p < 0.001$, uncorrected) during pain anticipation in women. During pain, placebo-induced down-regulation of the insula was altered in women compared to men (ROI analysis: $p_{FWE} < 0.05$). **Conclusions & Inferences** Our data provide first evidence supporting sex differences in pain-induced neural modulation during visceral placebo analgesia despite similar placebo-induced reductions in perceived pain intensity. These preliminary findings might contribute to elucidating mechanisms mediating placebo effects in clinical conditions associated with chronic abdominal pain such as in irritable bowel syndrome.

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INTRODUCTION

There is a widespread agreement about sex differences in the prevalence of chronic pain syndromes, which are overrepresented in women.^{1–3} Chronic abdominal pain or discomfort characterize the functional gastrointestinal disorders (FGDs), such as the irritable bowel syndrome (IBS), which constitute highly prevalent conditions with a marked female preponderance.⁴ Treatment options for these complex disorders remain limited, which is at least in part attributable to insufficient knowledge about the mechanisms mediating visceral hyperalgesia and central pain amplification – two phenomena that are considered crucial aspects in the pathophysiology of IBS and other FGDs.⁵ Despite calls for ‘personalized treatments’ in many medical fields, to date there is insufficient knowledge about the role of the patient’s sex and/or gender in the response to visceral pain in general and specifically in the response to treatment.

In the context of understanding and treating pain, it is crucial to consider psychological mechanisms, including the role of expectations. Evidence from the growing field of placebo research has convincingly underscored that any therapeutic intervention takes place in a psychosocial context that shapes patients’ expectations regarding a specific therapy, including the response to placebo treatment.⁶ The clinical relevance and therapeutic potential of placebo interventions has been demonstrated in many medical conditions, including IBS.⁷ Although placebo analgesia is arguably one of the most impressive and most frequently studied phenomena in the placebo field,⁶ knowledge regarding possible sex differences in placebo analgesia is scarce. Indeed, a recent review concluded that ‘studies are urgently required to better understand the role of sex-gender on placebo mechanisms and its impact on randomized clinical trials outcomes’.⁸ Even though there exists some evidence from experimental placebo studies and clinical trials focusing on somatic pain,^{9–11} nausea,^{12,13} and IBS symptoms¹⁴ supporting the existence of sex differences, it remains elusive which sex shows larger placebo responses and may hence be considered to be more ‘placebo-prone’. It is also unknown if findings in somatic pain^{9–11} or nausea^{12,13} models transfer to visceral pain, given significant differences in the peripheral and central pathways.¹⁵ Thus far, no study has specifically assessed if and to what extent sex differences in placebo responses to visceral pain, i.e.,

visceral placebo analgesia, exist. Hence, in this exploratory functional magnetic resonance imaging (fMRI) study in a relatively small sample, we compared healthy men and women with respect to behavioral and neural responses to rectal distensions in an established visceral placebo analgesia paradigm that induces positive expectations.^{16,17} Given conflicting data demonstrating greater placebo responses in men,^{9–11,18} in women,¹⁴ or no sex differences¹⁹ in other pain modalities and nausea, our analysis was explorative and had no directed hypotheses with respect to the magnitude of the placebo response or its underlying neural circuitry in men compared to women.

MATERIALS AND METHODS

Recruitment, inclusion and exclusion criteria

The recruitment and scheduling process included a structured telephone screening and an initial personal interview. Given that this study addressed effects of expectations on visceral pain, only subjects were recruited who were naïve with respect to all pertinent aspects of this study, including prior experience with rectal barostat distensions and prior use of the supposed painkiller (i.e., the spasmolytic drug Butylscopolaminiumbromid, which was in fact only saline). Of note, for the purpose of this analysis, we extended an existing placebo subgroup of a recently published study on placebo and nocebo responses.¹⁶ To do so, we additionally recruited $N = 18$ naïve, age-matched (± 3 years) healthy volunteers (9 women, 9 men) by public advertisement to provide a total sample of $N = 36$ healthy volunteers ($N = 18$ women; $N = 18$ men) matched by sex and age. Note that we herein report on full datasets available for $N = 30$ (15 women, 15 men), as explained below. Exclusion criteria were body mass index < 18 or > 30 , age < 18 or > 65 years, and any medical condition including gastrointestinal, neurological, psychiatric or endocrinological conditions. The Hospital Anxiety and Depression Inventory (HADS) was used as a screening tool for current anxiety or depression.²⁰ MR-related exclusion criteria consisted of claustrophobia, pregnancy, and ferromagnetic implantations. Symptoms suggestive of any functional or organic gastrointestinal condition were ruled out by interview and a standardized in-house questionnaire assessing a variety of typical upper and lower gastrointestinal symptoms, with scores ranging from 0 to 39.¹⁶ Sum scores ≥ 10 were exclusionary. Anorectal abnormalities were excluded by digital examination on the day of the study prior to balloon placement by a medical doctor. Structural brain abnormalities were excluded by structural MRI. Unless postmenopausal or on hormonal contraceptives, naturally cycling women were studied in the follicular phase of the cycle while avoiding the time of menstruation. A urine test was accomplished on the day of the study to rule out pregnancy. Ethics approval was granted by the University Hospital Essen ethics committee (permit no. 08-3823). Informed written consent was obtained during the initial personal visit and subjects were paid for their participation.

Study design and procedures

On the fMRI study day, rectal perceptual and pain thresholds were initially determined outside the scanner using a pressure-controlled barostat device as previously described in

detail.^{16,17,21–23} After a following short rest period (15 min), a structural MRI scan was acquired (5 min). To assess blood oxygen-level dependent (BOLD) responses during cued anticipation and pain delivery (i.e., rectal distensions), three identical, event-related fMRI sessions (7 min each) were accomplished. In all three sessions, subjects received rectal distensions at a pressure just below the individual pain threshold in order to establish similar perceptual intensities across individuals and between groups. In each session, eight cued distensions (duration: 16.8 s each) were delivered. The cue consisted of a visual signal (i.e., a blinking red cross, which stopped flashing during distensions) that appeared at pseudo-randomized intervals 2–5 scans prior to initiation of balloon distension. Resting periods ('off-phases'; duration: 16.8 s each) were announced by a different visual cue (i.e., a white cross). During scanning, subjects were prompted to rate each distension with respect to perceived pain intensity using visual analogue scales (VAS). Prior to each scanning session, volunteers completed additional VAS ratings of present tension and expected pain intensity for the subsequent session. Pauses in-between sessions were 20 min.

The first scanning session (session 1) served as an adaptation (or habituation) phase without intravenous (i.v.) drip in order to familiarize the naïve volunteers with the rectal stimuli and the MR-setting. This initial phase was also included to allow assessment of possible sex differences in this distension protocol irrespective of specific treatment expectations. Subsequent to the adaptation session, prior to initiation of sessions 2 and 3, subjects were informed about the sequence of upcoming treatments, and an indwelling i.v. catheter was placed in the left arm vein for continuous application of saline solution as i.v. drip. The same placebo intervention as utilized in our previous studies^{17,23,24} was implemented using deceptive instructions (for details, see below). In one session (session 2 or 3), participants were informed that they would receive a highly potent analgesic drug (100% expectation). In the other session (session 3 or 2, respectively), which served as a control condition, they were truthfully told that an inert substance, i.e. sodium chloride, was administered (0% expectation). The order of scanning sessions 2 and 3 was counterbalanced to control for sequence effects, i.e., half of the subjects underwent the 0% expectation condition first and vice versa.

Placebo instructions and procedures

As previously described in detail,^{16,17} instructions were standardized as much as possible and included both oral and written information provided as part of the informed consent visit and during the experiment. Instructions included detailed information on the supposed potent painkiller, i.e., the commercially available spasmolytic drug Butylscopolaminiumbromid, which subjects were told would be added to an i.v. saline drip during the experiment. Information included several statements on typical clinical uses of the substance and on about its known side effects. During the experiment, pertinent aspects of the instructions were repeated orally by the female investigators (J.S. or N.T.). Specifically, in the 100% condition, subjects were informed that they would receive the actual painkiller with the aim to analyze effects of this substance on neural responses to rectal pain. Ten minutes prior to the respective scanning session, a syringe clearly labeled with the drug name was injected into the container used for the continuous infusion drip in full view of the subjects. Subjects were again reminded about the nature and purpose of the treatment to ensure full attention, and were told that a waiting period of 10 minutes between drug administration and initiation of scanning was necessary to guarantee a sufficient drug effect. In

the 0% expectation condition, subjects were informed that an inert substance (i.e., sodium chloride; 'saline') without any effects on pain would be administered. They were told that the purpose of this condition was to control for all possible factors present in the complex MRI environment that could potentially influence neural responses to painful distensions, such as somatosensory responses to the i.v. drip itself. The container used for the infusion drip was clearly marked as 'sodium chloride'.

VAS ratings

During scanning, all visual stimuli were presented using commercially available stimulus delivery and experimental control software (Presentation®, Neurobehavioral Systems, Albany, CA, USA). VAS ratings were accomplished using a MRI-compatible response system (LUMItouch™, Photon Control Inc., Burnaby, BC, Canada). Using this system, subjects rated expected pain and present tension (prior to each session) and distension-induced pain (after each distension) on VAS scales, with ends defined as 'no pain/tension' and 'maximal pain/tension'. For analyses, all responses were quantified in mm from '0' to '100'. Pain ratings of all eight distensions per session were averaged for analysis.

Statistical analysis of non-fMRI data

Initially, normal distribution of the data was assured using Kolmogoroff–Smirnov test. Group comparisons of questionnaire data and of data from the adaptation phase were accomplished using independent-samples *t*-tests or chi-squared tests for dichotomous variables. Repeated VAS measures (expected and perceived pain; tension) were analyzed with repeated measures ANOVA with the factors group (women, men) and condition (0%, 100%; repeated factor). Note that *post hoc* testing was not accomplished given the absence of significant interaction or group effects. The alpha level for significance was set at $p < 0.05$. All non-fMRI data are shown as mean $\pm$ standard error of the mean.

fMRI imaging and analysis

Functional and structural images were acquired using a clinical 3 Tesla MR scanner (Magnetom Skyra, Siemens Health Care, Erlangen, Germany) with a 16 channel head coil. For structural imaging, a T1-weighted MPRAGE sequence with the following parameters was used: TR 1900 ms, TE 2.13 ms, flip angle 9°, FOV 239 mm, Matrix 256, slice thickness 0.9 mm, number of slices 192. Whole brain BOLD contrast images were acquired with a T2*-weighted gradient echo-planar imaging technique: TR 2400 ms, TE 26 ms, flip angle 90°, FOV 240 mm, Matrix 94, slice thickness 3 mm, number of slices 42. SPM 08 (Wellcome Trust Centre for Neuroimaging, London, UK) was used for further fMRI data analysis. Prior to statistical analysis, images were realigned to the mean, normalized to a standard EPI-template as implemented in SPM 08 and finally smoothed with an isotropic Gaussian kernel of 8 mm. Data were also subjected to high-pass filtering (cut-off period: 120 s), low-pass filtering with the hemodynamic response function and correction for temporal autocorrelations (based on a first-order autoregressive model). A voxel-by-voxel comparison according to the general linear model was used for 1st and 2nd-level analyses. First-level designs consisted of a box-car function convolved with the hemodynamic response function for two regressors (anticipation of pain; painful stimulation). Single-subject contrast images were entered into random effects group analyses, with the subject being the random

factor. Specific effects were tested with appropriate linear contrasts of the parameter estimates for the different regressors resulting in a *t*-statistic for each voxel. After model estimation, the ensuing 1st-level contrast images were used for 2nd-level (i.e., group) analyses treating individual subjects as a random factor and including nonsphericity correction.

We performed the following voxel-based analyses using one-sample *t*-tests and two-sample *t*-tests as implemented in SPM:

- (1) Initially, data from the adaptation phase (session 1) was analyzed with one-sample *t*-tests to confirm previous findings of rectal distension-induced neural activation. This was accomplished on the entire (pooled) sample ($N = 30$). Subsequently, we addressed sex differences in the neural response to rectal distensions in the adaptation phase (which may exist irrespective of specific treatment expectations). To accomplish this, two-sample *t*-tests contrasting anticipation and pain-induced neural responses, respectively, in men and women (i.e., men > women; men < women) were computed. As an additional, supplementary analysis, we compared the adaptation and 0% phases to test for possible novelty effects or other variability due to repeated measures but unrelated to expectations. Given statistical power limitations and the absence of sex-specific aims for this supplementary analysis, this was also carried out on the pooled sample ($N = 30$).
- (2) The main analysis addressing placebo-induced changes in neural activation was based on data from sessions 2 and 3, i.e., 100% expectation ('analgesic drug') compared to 0% expectation ('saline'), carried out in counterbalanced order. For each individual, differential contrasts were computed on the 1st-level, i.e., 0% > 100% and 100% > 0%. This was done separately for the pain anticipation phase and the pain delivery phase.
- (3) Subsequently, using these contrasts, the following 2nd-level analyses were carried out: We first conducted one-sample *t*-tests to describe placebo-induced changes in neural activation for pain anticipation and pain, respectively, separately for each group (i.e., within-group analyses).
- (4) To address the main question of sex differences in placebo responses, we conducted two-sample *t*-tests contrasting females > males and vice versa with respect to the differential contrasts computed on the first level, i.e., women_(100% > 0%) > men_(100% > 0%) and men_(100% > 0%) > women_(100% > 0%) together with the respective reverse contrasts.
- (5) Correlational analyses testing associations between placebo-induced changes in pain ratings (i.e., VAS pain reduction) and neural modulation were computed using Pearson's *r* based on peak voxel activation (with a sphere of 20 mm) for regions found significant in group comparisons (see point 4 above). This was done separately for men and women respectively.

We report results of voxel-based analyses using two different statistical thresholds: (1) Given the explorative nature of this study and comparatively small sample sizes, we initially conducted whole-brain analyses at $p < 0.01$ uncorrected. This is a relatively liberal approach which may lead to Type I error (i.e., spurious findings) and warrants careful interpretation. (2) To complement these findings with a more conservative method, we additionally computed regions-of-interest (ROI) analyses utilizing the established familywise-error (FWE) correction of alpha, reported as $p_{FWE} < 0.05$. The choice of ROIs was based on previous brain imaging findings in the fields of visceral pain and placebo analgesia.^{25–27} These included the insula (anterior; posterior), somatosensory cortices (S1, S2), thalamus, cingulate cortex, periaqueductal gray, dorsolateral prefrontal cortex (DLPFC), and ventrolateral prefrontal cortex (VLPFC). Except for the periaqueductal gray (sphere of 5 mm radius, anatomical center, $x = 3$, $y = -21$, $z = -3$, based on,²⁸ all ROI analyses were carried out

using anatomical templates constructed from the WFU Pick Atlas (Version 2.5.2), as implemented in SPM. All results are given as MNI coordinates.

RESULTS

Sociodemographic and psychological characteristics

Complete and valid questionnaire and behavioral data were available for $N = 18$ women and $N = 18$ men as per our recruitment goal. However, due to technical problems/movement artifacts, three male subjects had to be excluded. To assure comparability of behavioural and imaging data, we excluded the matching $N = 3$ women, and herein only report data for subjects with full behavioural and imaging datasets (i.e., $N = 15$ women, $N = 15$ men).

Women and men did not differ in age (confirming successful matching) or any other sociodemographic or psychological variables (Table 1). HADS anxiety and depression scores were comparable between women and men, and well-within the normal range. Furthermore, no sex differences were observed in self-reported minor gastrointestinal symptoms during the past 4 weeks or in rectal sensory or pain thresholds (Table 1).

Adaptation phase (session 1)

In the adaptation phase, ratings of expected pain (assessed prior to scanning: 53.1 ± 6.7 mm for women vs 36.3 ± 7.6 mm for men; $p = 0.10$) and of perceived pain intensity (assessed during scanning: 65.6 ± 4.7 mm for women vs 52.1 ± 6.0 mm for men, $p = 0.08$) tended to be higher in women when

Table 1 Participants

	Women ($N = 15$)	Men ($N = 15$)	p^*
Age (years)	36.9 ± 3.6	32.4 ± 2.8	0.33
Body mass index (kg/m^2)	23.4 ± 1.7	23.1 ± 0.7	0.84
Education ≥ 12 years % (N)	73.3 (11)	80.0 (12)	0.66
Smoker % (N)	20.0 (3)	20.0 (3)	1.00
HADS anxiety score	4.1 ± 0.4	4.1 ± 0.7	1.00
HADS depression score	1.5 ± 0.3	1.9 ± 0.5	0.47
Gastrointestinal symptom score [†]	3.3 ± 0.7	2.9 ± 0.6	0.71
Rectal sensory threshold (mmHg)	14.5 ± 1.2	13.6 ± 0.9	0.55
Rectal pain threshold (mmHg)	33.5 ± 2.4	32.7 ± 2.9	0.84

Data are shown as mean $\pm$ standard error or the mean, unless otherwise indicated. HADS, Hospital Anxiety and Depression scale; *Results of independent samples *t*-tests or chi-squared tests, values shown are exact *p*-values. [†]Sum score of a standardized in-house questionnaire assessing a variety of upper and lower gastrointestinal symptoms during the past month (range 0–39, cut-off ≥ 10).

compared to men. These differences reached significance when the full samples of $N = 18$ women and $N = 18$ men were compared (data not shown). No group differences were observed in tension ratings (women: 33.3 ± 6.9 mm vs men: 33.0 ± 7.8 mm, $p = 0.98$). The initial pooled analysis of BOLD responses during the adaptation phase (session 1), conducted on the pooled sample of $N = 30$ healthy males and females, essentially confirmed rectal distension-induced neural activation in this specific protocol. One-sample t -tests for anticipation and pain, respectively, revealed significant activations in a number of (expected) brain regions involved in the processing of visceral pain (Table 2). Upon two-sample t -tests comparing males and females for pain anticipation and pain, respectively, no evidence of sex differences was found in any region in the adaptation phase (data not shown). As a supplementary analysis, carried out on the pooled sample ($N = 30$), we

compared the adaptation and 0% sessions to test for possible novelty effects or other variability due to repeated measures that may occur irrespective of specific treatment expectations. The results revealed that in the adaptation session, neural activation was significantly greater in S2 both during pain anticipation (right S2: $x = 44$, $y = -12$, $z = 20$, $k_E = 21$, $t_{(p < .001 \text{ uncorr.})} = 4.39$; left S2: $x = -60$, $y = -18$, $z = 42$, $k_E = 43$, $t_{(p < .001 \text{ uncorr.})} = 4.1$) and during pain (right S2: $x = 46$, $y = -12$, $z = 22$, $k_E = 20$, $t_{(p < .001 \text{ uncorr.})} = 4.02$). The reverse contrasts did not result in significant activations.

Placebo effects (sessions 2 and 3)

In the 100% expectation condition, there was a trend toward decreased expected pain intensity (ANOVA condition effect: $F = 3.14$, $p = 0.09$, Fig. 1A) and

Table 2 Distention-induced BOLD responses during the adaptation phase ($N = 30$)

Contrast	Brain region	MNI coordinates				t -value	k_E	p
		H	x	y	z			
Pain	Insula	R	38	18	-2	10.59	1537	<0.05
		L	-38	10	0	7.30	406	<0.05
	ACC	R	4	16	44	6.05	38	<0.05
	Secondary somatosensory cortex	R	64	-30	32	6.71	92	<0.05
	Primary somatosensory cortex	R	56	-22	30	4.38	65	<0.05
		L	-60	-22	22	5.44	47	<0.05
Anticipation	DLPFC	R	30	48	22	6.39	63	<0.05
	DLPFC	L	-56	6	36	5.07	47	<0.05
	Secondary somatosensory cortex	R	52	6	8	4.77	151	<0.05
	Insula	R	34	18	0	4.62	177	<0.05
		L	-34	10	0	4.62	177	<0.05
		L	-56	6	36	5.07	47	<0.05

Distention-induced changes in neural activation over the whole sample ($N = 30$) during cued-pain anticipation and pain. Results of one-sample t -tests for pain (upper section) and anticipation (lower section) in regions-of-interest analyses (all $p < 0.05$, FWE corrected; results of whole-brain analyses not shown). H, hemisphere; DLPFC, dorsolateral prefrontal cortex; ACC, anterior cingulate cortex; k_E , cluster size.

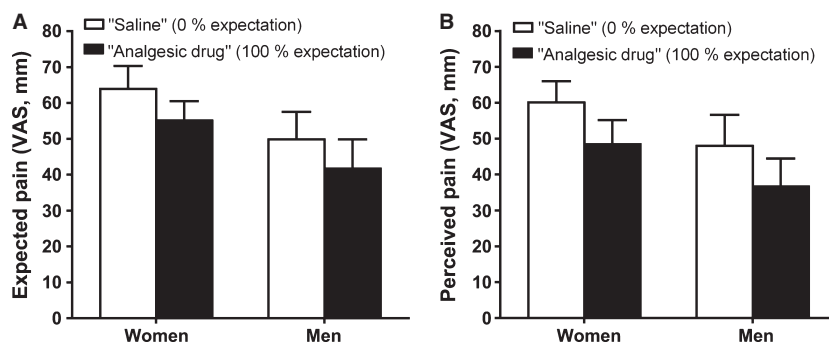


Figure 1 Visual analogue scale (VAS) ratings of expected pain (A) and perceived pain intensity (B) in $N = 15$ men and $N = 15$ women in a placebo condition (expectation to receive an analgesic drug: 100% expectation) and a control condition (expectation to receive saline: 0% expectation). In reality, subjects received saline solution in both conditions, which were carried out in counterbalanced order. Expected pain was assessed prior to each session. Perceived pain was rated during scanning after each distension, and ratings were averaged for analysis. In the 100% expectation condition, expected pain tended to be reduced (ANOVA condition effect: $p < 0.1$, A) and perceived pain was significantly reduced (ANOVA condition effect: $p < 0.05$, B). No significant group or condition $\times$ group effects were observed. For detailed ANOVA results, please see text. Data are shown as mean $\pm$ standard error of mean.

significantly reduced perceived pain intensity (ANOVA condition effect: $F = 5.15$, $p < 0.05$, Fig. 1B) when compared to the 0% expectation condition. No sex differences in these responses were observed, as indicated by a lack of significant interaction effects (for expected pain: $F = 0.006$, $p = 0.94$; for perceived pain: $F = 0.001$, $p = 0.98$), or main effects of sex (for expected pain: $F = 2.5$, $p = 0.13$; for perceived pain: $F = 1.74$, $p = 0.20$). No effects were observed for tension ratings (all $p > 0.1$, data not shown).

To assess the neural mechanisms mediating visceral placebo analgesia, neural activation in the 0% expectation condition ('saline') was compared with the 100% expectation condition ('analgesic drug') utilizing the differential first-level contrasts for pain anticipation and pain respectively (i.e., 0% > 100%; 0% < 100%). Initial within-group analyses with one-sample t -tests (Table 3) revealed that within the group of women, the 100% expectation condition ('analgesic drug') led to significantly greater activation of the VLPFC and anterior insula (both $p_{\text{FWE}} < 0.05$), and additionally of the DLPFC and superior parietal gyrus (whole-brain analysis at $p < 0.001$, uncorrected) during pain anticipation. During painful distensions, women revealed

increased activation of the DLPFC, ACC and putamen in the 100% condition (whole-brain analyses: all $p < 0.001$, uncorrected). The reverse contrast (i.e., 0% > 100%) revealed activation in the superior temporal gyrus within women. Within the group of men, the 100% expectation condition ('analgesic drug') led to reduced activation of the insula ($p_{\text{FWE}} < 0.05$) and additionally of the middle temporal gyrus and posterior cingulate cortex in whole-brain analyses (all $p < 0.001$, uncorrected) during pain anticipation. No significant results were found for pain within men.

To directly test sex differences in BOLD responses during placebo analgesia, two-sample t -tests on the differential 1st-level contrasts (i.e., 100% > 0%; 100% < 0%) were conducted (Table 4). Group differences upon two-sample t -tests are shown in Fig. 2; for illustration purposes and to allow proper interpretation of the directionality of findings in these differential contrasts, see Fig. 3 for contrast estimates. In the 100% condition compared to the 0% condition, women demonstrated greater modulation of the insula (Figs 2A and 3A) and DLPFC (Figs 2A and 3B) during cued pain anticipation ($p_{\text{FWE}} < 0.05$ for posterior insula; all other $p < 0.001$, uncorrected). During pain, placebo-induced

Table 3 Placebo-induced changes in neural activation within each group ($N = 15$ women; $N = 15$ men)

		MNI coordinates						
Group/Contrast	Phase/Brain region	H	x	y	z	t-value	k _E	p
Women								
0% < 100%	Pain anticipation							
	VLPFC	R	65	18	4	6.40	117	0.017
	DLPFC	R	48	16	18	4.34	42	<0.001
	Insula (anterior)	R	42	20	-8	4.64	122	0.037
		L	-48	0	-6	4.26	24	<0.001
	Superior parietal gyrus	L	-28	-52	0	4.13	6	<0.001
	Pain							
	DLPFC	R	26	40	20	4.43	21	<0.001
	ACC	R	10	12	28	4.22	12	<0.001
0% > 100%	Putamen	R	22	14	-14	3.28	6	<0.001
	Pain anticipation							
	-							
	Pain							
	Superior temporal gyrus	L	-66	-42	14	4.32	16	<0.001
Men								
0% < 100%	-							
0% > 100%	Pain anticipation							
	Insula	R	44	0	18	6.20	66	0.025
	Middle temporal gyrus	L	-52	-72	28	4.66	16	<0.001
	PCC	L	-10	-46	10	4.30	7	<0.001
	Pain							
	-							

Placebo-induced changes in neural activation within each group of $N = 15$ women (upper section) and $N = 15$ men (lower section) during cued-pain anticipation and pain contrasting 100% expectation ('analgesic drug') with 0% expectation ('saline') and vice versa. Results of whole-brain analyses (all $p < 0.001$, uncorrected). Activations which reached significance in additional, more conservative ROI-analyses with FWE correction of alpha (i.e., VLPFC, insula) are highlighted in bold font and p -values shown are exact p -values. '–' indicates no significant findings in either the whole-brain or the ROI-analysis. H, hemisphere; DLPFC, dorsolateral prefrontal cortex; VLPFC, ventrolateral prefrontal cortex; ACC, anterior cingulate cortex; PCC, posterior cingulate cortex; k_E , cluster size.

Table 4 Group differences in placebo-induced changes in neural activation between women ($N = 15$) and men ($N = 15$)

Contrast	Phase/Brain region	MNI coordinates				<i>t</i> -value	k_E	<i>p</i>
		H	<i>x</i>	<i>y</i>	<i>z</i>			
Women > Men 100% > 0%	Pain anticipation							
	Insula (posterior)	R	42	-10	8	3.86	28	0.04
	Insula (anterior)	R	34	20	-6	3.36	104	0.001
	DLPFC	R	46	30	14	3.75	27	<0.001
	Pain							
	Insula (anterior)	R	40	4	-2	3.98	22	0.03
	Insula (posterior)	R	40	-2	16	3.85	51	<0.001

Group differences in placebo-induced changes in neural activation in $N = 15$ women compared to $N = 15$ men during pain anticipation and pain. Results of two-sample *t*-tests comparing the groups (women > men, upper section; women < men, lower section) with respect to the first-level differential contrasts between the 100% expectation condition ('analgesic drug') and the 0% expectation condition ('saline'). Results of whole-brain analyses (all $p < 0.001$, uncorrected). Activations which reached significance in additional ROI-analyses with FWE correction of alpha (i.e., modulation of the insula during anticipation and pain) are highlighted in bold font and *p*-values shown are exact *p*-values. Other contrasts revealed no significant activations either in whole-brain or ROI-analyses (data not shown). H, hemisphere; DLPFC, dorsolateral prefrontal cortex; k_E , cluster size.

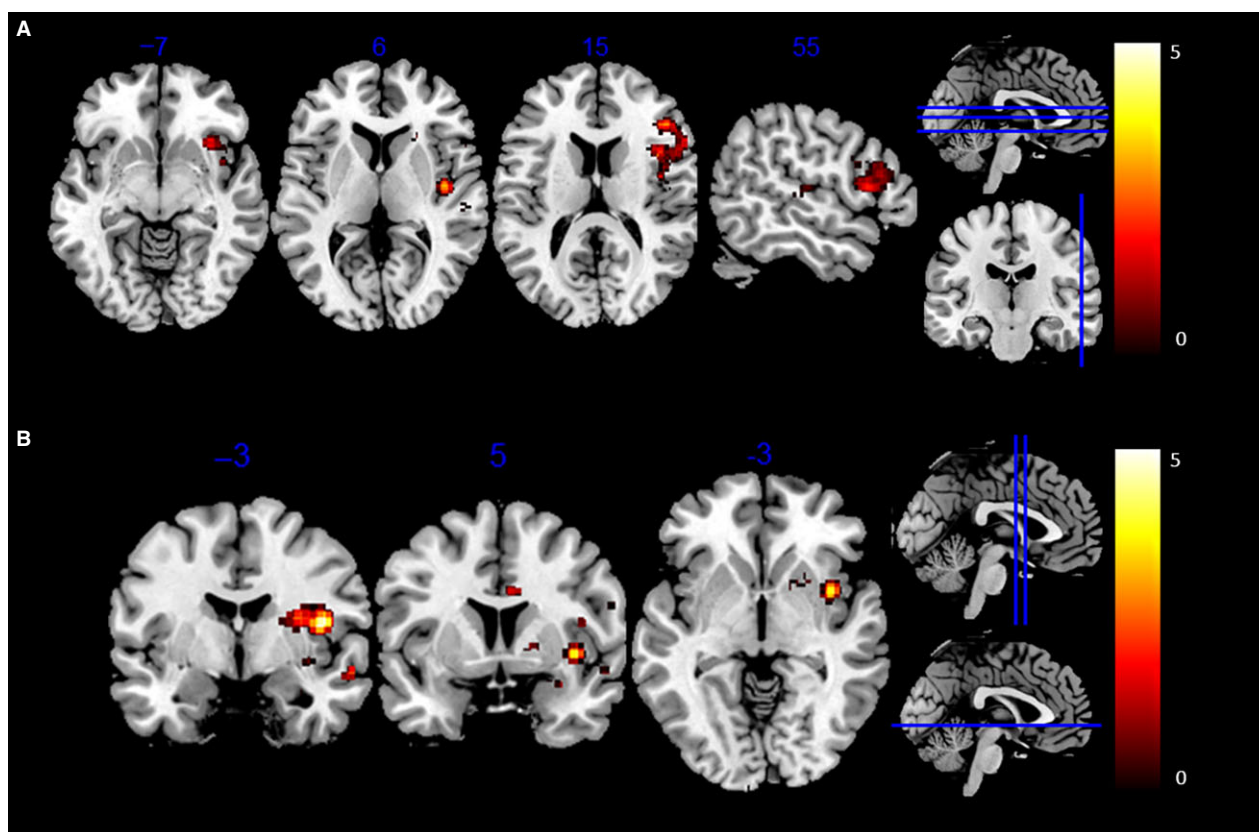


Figure 2 Results of two-sample *t*-tests comparing placebo-induced neural modulation in $N = 15$ women and $N = 15$ men during cued pain anticipation (A) and painful stimulation (B). (A) Women showed greater modulation of the posterior insula (ROI analysis: $p_{FWE} < 0.05$) and dorsolateral prefrontal cortex (DLPFC, whole-brain analysis: $p < 0.001$, uncorrected) during cued pain anticipation. (B) During pain, placebo-induced down-regulation of the insula was altered in women compared to men (ROI analysis: $p_{FWE} < 0.05$). Images overlaid on a structural T_1 -weighted MRI used for spatial normalization and thresholded at $p < 0.01$ uncorrected using anatomical masks for visualization purposes. Color bars indicate *t*-scores. For coordinates, exact *t*-scores and cluster sizes, see Table 4. See Fig. 3 for plots of contrast estimates for peak-voxel activations for these contrasts and groups.

down-regulation of the insula was altered (i.e., absent) in women compared to men who revealed a downregulation ($p_{FWE} < 0.05$, Figs 2B and 3C). Correlational

analyses were computed to test correlations between changes in subjective pain (VAS pain reduction = magnitude of the placebo response) and changes in

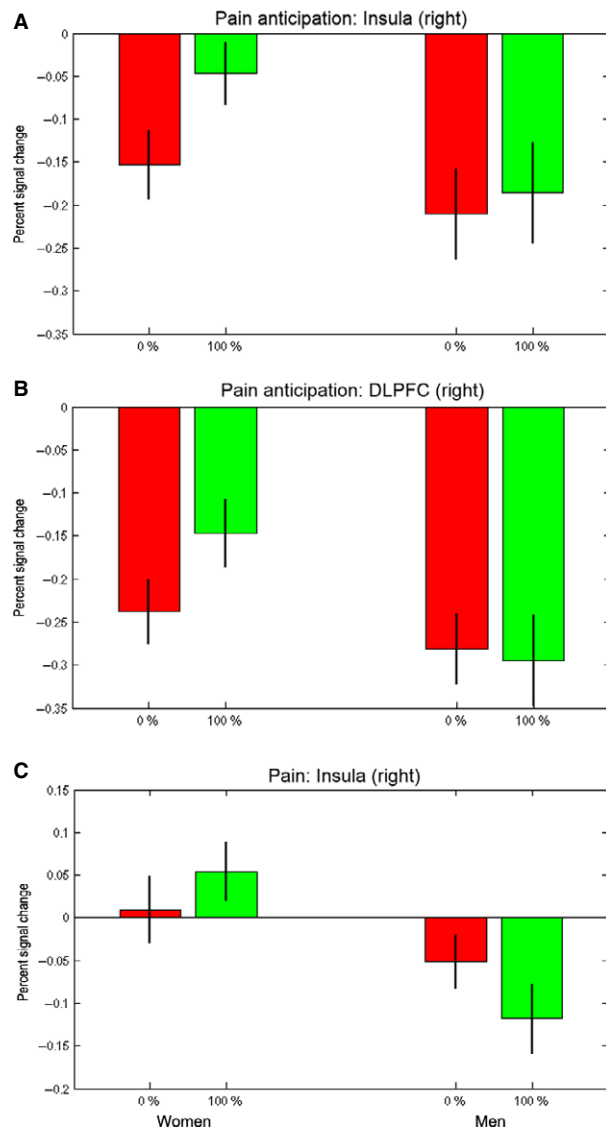


Figure 3 Plots of contrast estimates of peak voxel activation for regions found significant upon two-sample tests comparing $N = 15$ men and $N = 15$ women with respect to placebo-induced changes during cued pain anticipation (A: insula; B: dorsolateral prefrontal cortex) and painful stimulation (C: insula). Neural activation was assessed in a placebo condition (green bars: 100% expectation induced by deceptive instructions of receiving a spasmolytic drug) and in a control condition (red bars: 0% expectation induced by truthful instructions of receiving saline), carried out in counterbalanced order. Compare to Fig. 2 and Table 4, a.u., arbitrary units.

neural modulation (change in peak voxel activation). These supplementary analyses, carried out separately for men and women, revealed significant negative correlations between the magnitude of placebo response and change in placebo-induced neural activation only within women for the DLPFC ($r = -0.55$, $p < 0.05$) and insula ($r = -0.43$, $p < 0.05$) during pain anticipation.

DISCUSSION

Existing data on sex differences in placebo responses are sparse and conflicting, revealing greater placebo responses in men,^{9–11,18} in women,¹⁴ or no sex differences.¹⁹ Our study is the first to address sex differences in placebo analgesia in a visceral pain model. To do so, we compared healthy men and women with respect to the behavioral and neural responses to painful rectal distensions in an established placebo analgesia paradigm. In summary, women and men showed comparable placebo-induced reductions in pain ratings. Analysis of fMRI data, on the other hand, was indicative of sex differences in the neural mechanisms mediating placebo analgesia, which is interesting in the context of understanding the central mechanisms mediating treatment responses for acute and chronic visceral pain in conditions such as IBS.

Sex differences in neural modulation during placebo analgesia

Analysis of fMRI data revealed evidence suggestive of differences between men and women in placebo-induced neural modulation. Of note, as this constitutes the first fMRI study addressing sex differences in placebo analgesia, we conducted an exploratory analysis using (relatively liberal) whole-brain analyses with a threshold of $p < 0.001$ (uncorrected). The results of these analyses have to be interpreted with caution. To complement these results with a more conservative method of alpha correction (and hence a reduced probability of spurious findings), we also provide results of ROI analyses with alpha correction (at $p_{FWE} < 0.05$). In women, we observed increased placebo-induced modulation of the posterior insula (at $p_{FWE} < 0.05$) and of the DLPFC (at $p < 0.001$ uncorrected) during cued pain anticipation. This enhanced anticipatory modulation may reflect sex differences in the integration of anticipated nociceptive input and altered cognitive appraisal processes in the context of preparing for a painful experience. During pain, women demonstrated significantly less placebo-induced down-regulation of the insula (at $p_{FWE} < 0.05$) when compared to men (in fact, as suggested by contrast estimates reflecting changes in peak voxel activation, women actually revealed a slight up-regulation). The insula is of primary importance for interoception and multi-modal sensory integration in the context of pain^{29,30} and plays a role in pain-related decision-making,³⁰ emotional awareness, and the integration of (anticipated) interoceptive stimuli.^{29,31} Hence, down-regulation of pain-induced

insula activation during placebo analgesia in men may reflect the ability to 'dampen' painful afferent sensory information to reduce pain-related emotional and cognitive processing at the cortical level. In other words, women appear to be less capable of this cognitively driven neural down-regulation of insular activation during placebo analgesia. This notion is supported by supplementary correlational analyses suggesting a negative association between the magnitude of placebo-induced changes in peak voxel activation for the DLPFC and insula during pain anticipation and the behavioral placebo response (i.e., reduction in VAS pain rating). In other words, greater neural anticipatory response was associated with a poorer pain reduction only in women. Of note, the differential neural processing of visceral placebo analgesia was not related to sex differences in pain processing *per se*, as neural responses to pain anticipation and painful stimulation did not differ significantly between men and women, in line with our previous report in a different sample of healthy volunteers.²² Together, these findings are interesting in light of recent evidence that IBS patients reveal disturbed neural modulation during placebo analgesia.^{16,25} Clearly, future studies should address the putative role of sex and/or gender in altered cognitively driven central pain inhibition in patient populations with chronic abdominal pain such as IBS.

Placebo effects at the subjective level (ratings)

As evidenced by significant reductions in ratings of perceived pain in the placebo condition, the placebo paradigm utilized herein was clearly effective, which is in line with our own previous data^{16,17,23,24} and results from other groups utilizing similar experimental approaches to induce placebo analgesia in visceral pain models.^{25,26,32–35} Interestingly, the behavioral placebo effects observed herein were comparable between men and women, despite initially enhanced pain expectation and perceived pain ratings in the adaptation phase in women. Hence, our data support that although women may show altered subjective pain evaluation *per se*, their behavioral response to a suggestion-based placebo intervention is unaltered when compared to men. The present negative results for sex differences in placebo analgesia at the level of pain ratings are consistent with placebo data from another clinical pain model, i.e., the molar extraction dental pain model.¹⁹ At the same time, they are at odds with placebo studies utilizing somatic pain models, including heat pain,^{9,36} cold pain,¹⁸ electrically induced pain¹¹, or tourniquet technique,¹⁰ which

were all in support of more pronounced placebo responses in men. Given our visceral pain model and a previous study showing greater placebo responses in female IBS patients,¹⁴ one may speculate that effects of sex on the subjective evaluation of pain during placebo analgesia may depend on pain modality and/or chronic pain condition. Clearly, in addition to organ system and/or patient status, another putative source of variation may lie in the underlying placebo mechanism, i.e. expectation (as modeled herein) vs learning/conditioning (not addressed herein).¹² Either way, even in the absence of obvious sex differences at the behavioral level, our VAS data further underscore that expectation-based placebo interventions effectively reduce visceral pain both in healthy men *and* women, irrespective of overall pain expectation and responsiveness, which has implications for the treatment of conditions associated with chronic abdominal pain. Whether or not the underlying brain circuitry actually differs according to sex needs to be further clarified in studies with larger samples.

Critical considerations and limitations

There exist several critical aspects worth of discussion. Firstly, the sex of the experimenter interacting with the participants was standardized (herein: female investigators), but not systematically varied. Hence, we cannot exclude possible interaction effects between the sex of the investigator and the sex of the subject, which reportedly plays a role at least in different somatic pain models.^{36,37} For example, in a heat pain experiment, Aslaksen *et al.*³⁶ reported that investigator × subject sex interaction influenced pain ratings with lower pain reports in male subjects given female investigators, whereas changes in autonomic parameters remained unaffected. A similar interaction process may have contributed to our finding of lower pain reports in men compared to women in the adaptation phase. On the other hand, our fMRI data are less likely affected by investigator × subject sex interactions, given the highly standardized, electronic rating system (for VAS pain ratings) during the scanning session without direct interaction with the staff. Furthermore, we introduced an adaptation session to reduce possible effects of novelty (which we indeed observed for S2 in a supplementary analysis), and counterbalanced placebo and control sessions afterwards, all of which are strengths of this study design. Secondly, data acquisition and analysis were not blinded, which may have introduced a bias. Thirdly, we focused on placebo analgesia in a paradigm that modifies

expectations (i.e., that is based on positive suggestions). In addition to expectation, placebo effects are also mediated by learning/conditioning processes, which we did not address. However, it will be important to extend our findings to also address possible sex differences in placebo analgesia based on conditioning protocols, as this may differentially influence placebo responses in men and women as previously demonstrated.¹² Finally, our results must be interpreted in the light of the small samples which bears the risk of type II errors, i.e., of an underestimation of possible sex differences.

CONCLUSIONS

In summary, women and men demonstrated different placebo-induced neural modulation during cued pain activation and delivery of noxious rectal stimulation involving the insula and DLPFC. At the same time, women and men showed comparable placebo response at the level of subjective pain evaluation. Clearly, these results require further evaluation in larger healthy and patient populations, but provide first evidence for sex-differences in the neural activation in response to a placebo intervention.

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DISCLOSURE

All authors declare no conflict of interest.

AUTHOR CONTRIBUTION

NT performed the research, analyzed data, and drafted the manuscript; JS performed the research and analyzed data; AI performed the research; CM performed the research; MF designed the study and acquired funding; ERG designed the study and acquired funding; MS designed the study and acquired funding; SE designed the study, acquired funding, and wrote the final version of the manuscript; SB analyzed data and wrote the final version of the manuscript. All authors contributed to the interpretation of the data, revised the manuscript, and approved the final version of the manuscript.

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